YEAR 7 · MATHS UNIT 5 · SCIENCE UNIT 3 · HOMEWORK 11

# 180° Apart

Part 1: Maths — 5.1 A sum of 360° and 5.2 Intersecting lines
Part 2: Science — 3.4 Tides, 3.5 Energy and 3.6 Changes in energy
Part 3: Maths and science together

**Name:** _____ **Class:** _____

**Date given:** _____ **Date due:** Friday 6 November 2026

**How to do this homework**

- **A–B:** maths. **C–E:** science. **F:** both together. **No calculator.** Write the **calculation** first, then the answer.
- **Give a reason** = write the angle fact after **because**: *"a = 40° because they are vertically opposite angles."*
- Answer in **full sentences** when it says so. *"Kinetic"* is not a sentence. *"The bike has kinetic energy because it is moving"* is.

## Section A · Maths 5.1 — A Sum of 360°

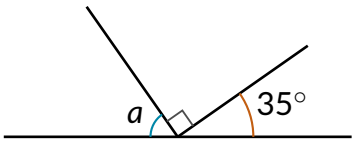
### A1 Find the missing angle: two steps

**Two steps, from the lesson**

**1. Add** the angles you know. **2. Subtract** the total from **180°** (a straight line) or **360°** (around a point, or in a quadrilateral).
In a **parallelogram**, opposite angles are **equal**.

The diagrams are drawn accurately. Find each letter. (a) is done for you.

**(a)**

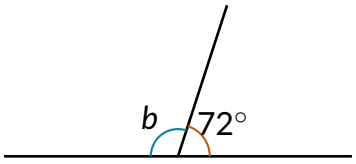



**1.** 35° + 90° = 125°

**2.** 180° − 125° = 55°

**a** = 55°

**(b)**

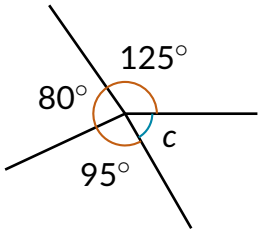



**1.** _____

**2.** _____

**b** = _____

**(c)**

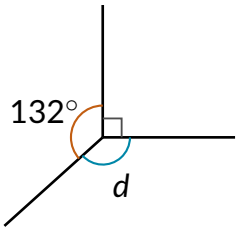



**1.** _____

**2.** _____

**c** = _____

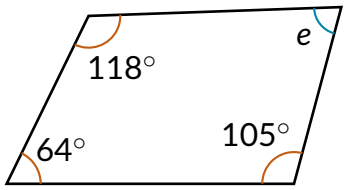
**(d)**




1. _____
2. _____

**d** = _____

**(e)**




1. _____
2. _____

**e** = _____

**(f)**




a parallelogram

**q** = _____

1. _____

**p** = **r** = _____

## A2 From English to a calculation

**Word help — the words in angle questions**

| | | | |
|---|---|---|---|
| **sum** | the total when you add | **work out** | find, and show the calculation |
| **equal** | the same size | **a full turn** | 360° |

Write the **calculation first**. Then write the answer.

(a) Two angles are 63° and 116°. Mr Bowen says they make a **straight line**. Is he right?

**Calculation:** _____ **Answer:** _____

(b) The sum of two angles is 180°. One of them is 64°. Work out the other one.

**Calculation:** _____ **Answer:** _____

(c) Three angles of a quadrilateral are 90°, 90° and 47°. Calculate the fourth angle.

**Calculation:** _____ **Answer:** _____

(d) One angle of a quadrilateral is 120°. The other three angles are equal. Work out the size of each one.

**Calculation:** _____ **Answer:** _____

(e) Mr Bowen spins round on his chair. He turns through 1080°. How many full turns is that?

**Calculation:** _____ **Answer:** _____

## A3 Now you write the question

Write a word problem for $360° - 70° - 110° - 95° = 85°$

**Your word problem must have: tick each one**

- ☐ a real thing (a garden, a kite …)
- ☐ the word **quadrilateral**
- ☐ three angles
- ☐ full sentences
- ☐ a question at the end
- ☐ the answer 85°

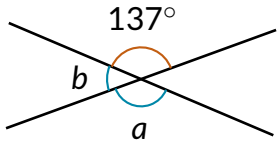
## Section B · Maths 5.2 — Intersecting Lines

## B1 Find the angle. Give a reason.

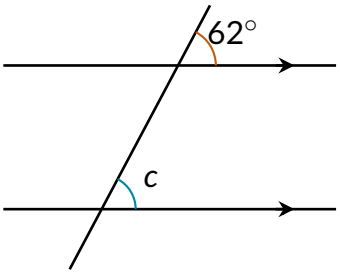
**The four reasons — copy the one you need after because**

**X:** vertically opposite angles are equal
**F:** corresponding angles are equal
**line:** angles on a straight line add up to 180°
**Z:** alternate angles are equal
**Example:** *a* = 137° because vertically opposite angles are equal.

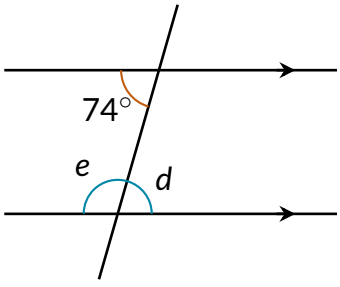
Arrows (>) on two lines mean they are **parallel**. Find each letter, and give a reason.




**(i)** two lines cross




**(ii)** parallel lines



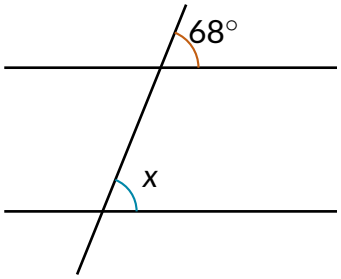

**(iii)** parallel lines

*b* = _____ because _____

*c* = _____ because _____

*d* = _____ because _____

*e* = _____ because _____




**(iv)** Which statement is correct: **A**, **B** or **C**?

**A** $x = 68°$

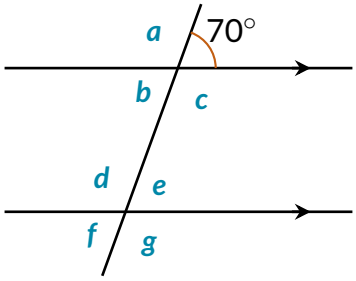
**B** $x = 112°$

**C** You need more information.

**Answer:** _____ because _____

_____

## B2 Mr Bowen's homework




Only **one** of Mr Bowen's sentences is completely right. Circle **right** or **wrong**. If it is wrong, write the correct sentence.

**(a)** *b* = 70° *because they are corresponding angles.* right · wrong

_____

**(b)** *e* = 70° *because they are alternate angles.* right · wrong

_____

**(c)** *a* = 70° *because vertically opposite angles are equal.* right · wrong

_____

**(d)** *d* = 110° *because d and c are alternate angles.* right · wrong

_____

# Section C · Science 3.4 — Tides

## C1 A day at the harbour

Mr Bowen writes down the tides at his harbour for one day.

| **Time** | 3 a.m. | 9 a.m. | 3 p.m. | 9 p.m. |
|---|---|---|---|---|
| **Tide** | high tide | low tide | high tide | low tide |
| **Depth** | 5.7 m | 1.3 m | 5.7 m | 1.3 m |

(a) How many high tides are there in one day? _____

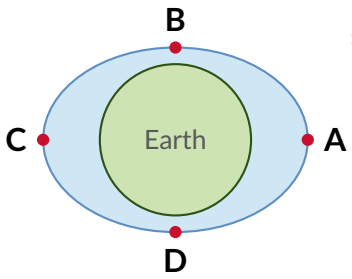
(b) How many hours is it from high tide to low tide? _____

(c) Work out the **tidal range**. (**tidal range**: the difference in depth between high tide and low tide)

**Calculation:** _____ **Answer:** _____

(d) Tomorrow, will the first high tide be at 3 a.m. again? Answer in a full sentence. Start: **"No, because the Moon …"**

## C2 Two high tides at the same time




seen from above the North Pole

not to scale




**(a)** Is each place at **high tide** or **low tide**? Write **H** or **L**.

**A** ☐ **B** ☐ **C** ☐ **D** ☐

**(b)** The Moon is on the right. Why is there a high tide at **C**, on the far side? Start: **"The near side is pulled the most. The far side …"**

## C3 The Sun, and the land

**(a)** The Sun's gravity is **stronger** than the Moon's. Why does the Moon make **bigger** tides? Start: **"The Sun is much further away, so …"**

**(b)** The land in Hanoi goes up and down about **30 cm**, twice a day. Why do we **not** feel it? Start: **"We do not feel it because …"**

# Section D · Science 3.5 — Energy

## D1 Mr Bowen's afternoon in joules

**The joules, from the lesson**

1 floor of stairs: **2 000 J** · run 1 metre: **200 J** · boil 1 litre of water: **400 000 J**

Write the **calculation first**. Then write the answer in joules (J).

(a) He climbs 4 floors to the science lab.

**Calculation:** _____ **Answer:** _____

(b) He runs 45 m to catch the bus. He misses it.

**Calculation:** _____ **Answer:** _____

(c) He boils 0.5 litres of water for his tea.

**Calculation:** _____ **Answer:** _____

(d) How many **times as much** energy does the tea use as the stairs? (**how many times as much**: divide)

**Calculation:** _____ **Answer:** _____

## D2 Mr Bowen's bike ride at night

Name the energy in each clue. Then write **store** or **transfer**. Row 1 is done for you.

| Clue | Energy | Store or transfer? |
|---|---|---|
| 1. The bike is moving. | kinetic | store |
| 2. Mr Bowen is at the top of a hill. | | |
| 3. The banana in his bag. | | |
| 4. The bike lamp shines. | | |
| 5. A current flows in the wire to the lamp. | | |
| 6. He rings the bell. | | |
| 7. He brakes. The brakes get hot. | | |
| 8. The tyres are squashed. | | |

## D3 Which stores last?

Will it keep its energy for a **year**? Write **Y** (yes) or **N** (no).

a cup of hot tea ☐ a packet of rice ☐ a spinning top ☐ a rock at the top of a cliff ☐

# Section E · Science 3.6 — Changes in Energy

## E1 An event or a process?

An **event** happens once. A **process** keeps going. Write **E** (event) or **P** (process) in each box.

a plate falls and breaks ☐ a balloon pops ☐ a candle burns ☐

the Sun shines ☐ a firework explodes ☐ a fridge keeps food cold ☐

## E2 Energy-change diagrams

Write the missing energy in each empty box. Read each arrow as **to**.

**(a)** A toaster




**(b)** Charging a phone



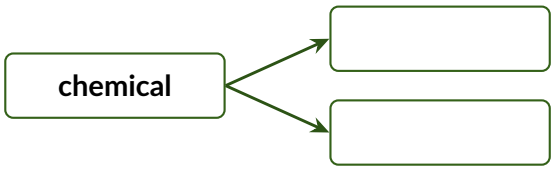

**(c)** A skateboard rolls down a ramp




**(d)** Mr Bowen rubs his hands



**(e)** A candle burns. Two arrows from one box.




## E3 The bouncing football

Mr Bowen drops a football from **2 m**. It bounces up to **1.3 m**.

**(a)** Which store is the **biggest** at each moment? Use: **kinetic** · **elastic potential** · **gravitational potential**

| Moment | Biggest store |
|---|---|
| in his hand, 2 m up | gravitational potential |
| just before it hits the floor | |
| squashed on the floor | |
| at the top of the bounce | |

(b) How much lower than 2 m does it bounce?

**Calculation:** _____ **Answer:** _____

(c) Why does it not bounce back up to 2 m? Answer in a full sentence. Start: **"At the bounce, some energy …"**

## E4 Hotter to colder

Mr Bowen's classroom is at 30 °C. Does thermal energy move **into** each drink, or **out of** it? Circle one.

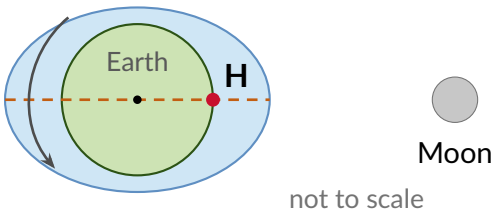
**(a)** a hot coffee at 80 °C **into** · **out of**

**(b)** a cold can of juice at 4 °C **into** · **out of**

**(c)** Mr Bowen opens the fridge. He says: *"Close the door! You are letting the cold out!"* Is he right? Explain. Start: **"He is …, because thermal energy …"**

# Section F · Maths and Science Together

## F1 180° apart




The Earth seen from above the North Pole. The two bulges are on a **straight line** through the middle of the Earth. Mr Bowen's harbour **H** has high tide now.

The Earth **spins** once a day: **360° in 24 hours**.

**(a)** How many degrees does the Earth turn in **1 hour**?

**Calculation:** _____ **Answer:** _____

**(b)** Complete the table.

| **Hours** | 1 | 3 | 6 | 12 | 24 |
|---|---|---|---|---|---|
| **Degrees** | 15° | | | | 360° |

(c) After **6 hours**, H has turned through a **right angle**. Is H at high tide or low tide then?

_____

(d) After **12 hours**, H has turned 180°. It is on the other side of the Earth. Is H at high tide or low tide?

_____

(e) Complete: The two bulges are _____° apart. So H has a high tide every _____ hours.

(f) Every day the Moon moves on a little. So the Earth must turn about **12° more** to bring H back to the bulge. The Earth turns 1° every **4 minutes**. How many minutes later is high tide each day?

**Calculation:** _____ **Answer:** _____

## F2 In a line, or at right angles?




**(i)** full moon **(ii)** half moon

seen from above the North Pole, not to scale

**(a)** Write the angle **?** at the Earth, between the line to the Sun and the line to the Moon.

**(i)** _____ **(ii)** _____

**(b)** In **(ii)**, the two lines are **parallel** · **perpendicular**. Circle one.

(c) In the lesson's model, the tidal range at Mr Bowen's harbour was **6 m** in one picture and **2 m** in the other. Which picture had the 6 m tides? _____

(d) How much **bigger** is the 6 m range? How many **times as big**?

**Bigger by:** _____ **Times as big:** _____

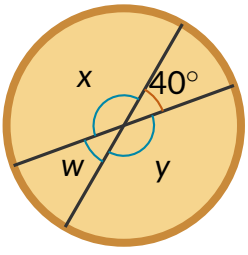
(e) Why are the tides biggest when the Sun, the Moon and the Earth are in a line? Start: **"When they are in a line, the Sun's tidal force …"**

## F3 A slice of energy

Food has a **chemical** store. Mr Bowen's pizza stores **3 600 000 J**. So every **1°** of pizza stores **10 000 J**.

(a) He cuts it into **8 equal slices**. What angle is one slice? How many joules does one slice store?

**Angle:** _____ **Joules:** _____




**(b)** The next pizza: he makes **two straight cuts**. One slice is 40°. Find *w*, *x* and *y*. Give a reason for each.

*w* = _____ because _____

*x* = _____ because _____

*y* = _____ because _____

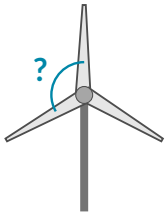
(c) He eats the **two biggest** slices. How many degrees of pizza is that? How many joules?

**Degrees:** _____ **Joules:** _____

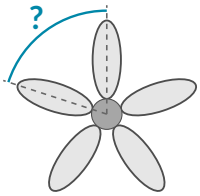
(d) One floor of stairs uses **2 000 J**. How many floors could he climb with the energy in those two slices? (Landmark 81 has 81 floors.)

**Calculation:** _____ **Answer:** _____

## F4 The wind turbine and the fan




a **wind turbine**: 3 blades




a **ceiling fan**: 5 blades

**(a)** The blades are **equally** spaced. Work out each angle **?**

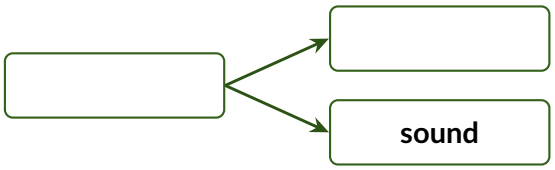
**Turbine:** _____ **Fan:** _____

**(b)** Fill in the energy-change diagrams. Use: **kinetic** · **electrical** · **sound**

**Turbine**



**Fan**




**(c)** What do you notice about the two diagrams? Answer in a full sentence. Start: **"The fan's energy change is …"**

## F5 Explain it with maths

Mr Bowen says: *"The Moon is on one side of the Earth. So there is only **one** high tide a day."* Write him a reply in English: 4 or 5 sentences.

**Your reply must have: tick each one**

| | | |
|---|---|---|
| ☐ **tidal force** | ☐ **two bulges** | ☐ **180°** apart |
| ☐ the Earth **spins** | ☐ **12 hours** | ☐ full sentences |